



Location & Telemetry Error

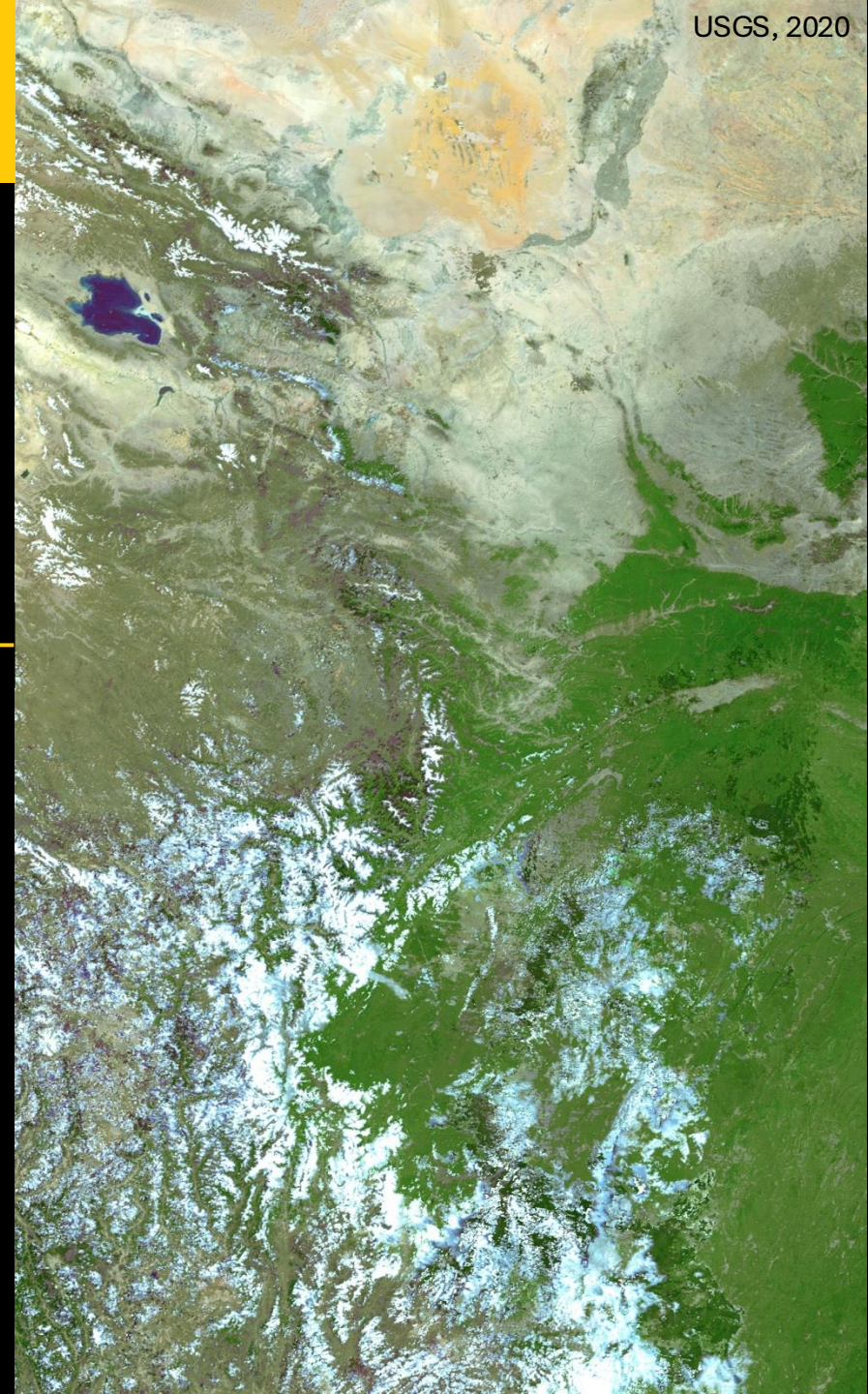
Continuous-Time Animal Movement Modeling

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Overview

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- 1) Introduction
- 2) Estimating error and building the error model
- 3) Implementation in `ctmm`
- 4) Error model selection

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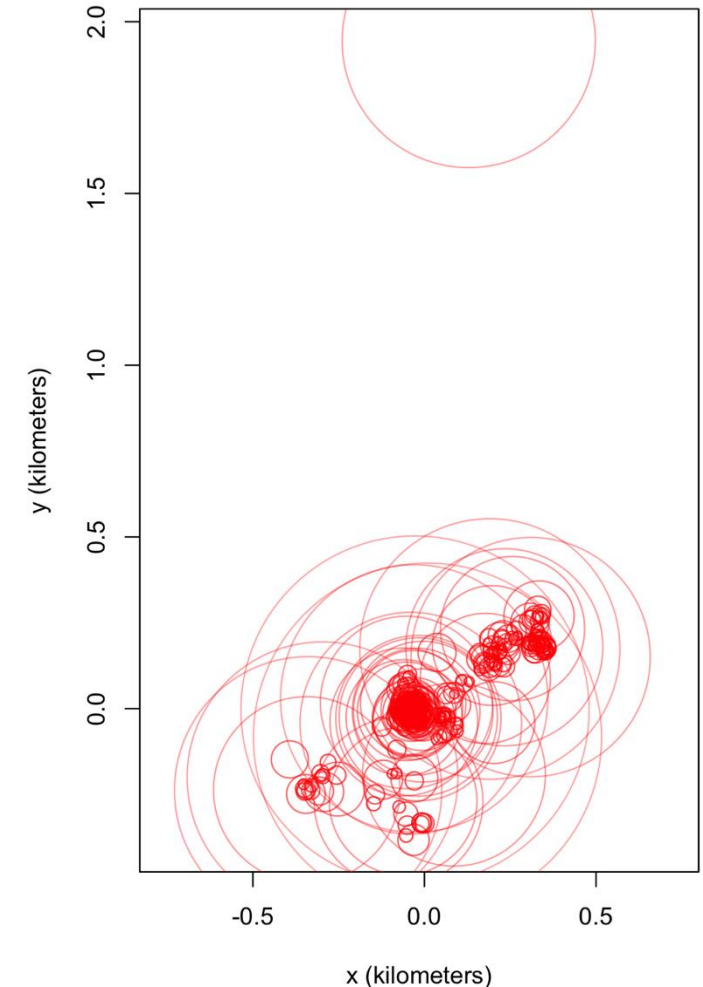
Introduction

- Location data are often shown as points
 - However, there is uncertainty associated with recorded locations
- Telemetry/location error (satellites) can impact the accuracy of our home range and speed/distance analyses
 - Consider an animal that does not travel much

EN



Wood turtle
(*Glyptemys insculpta*)



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 - How do the scales of movement compare to the scales of error?

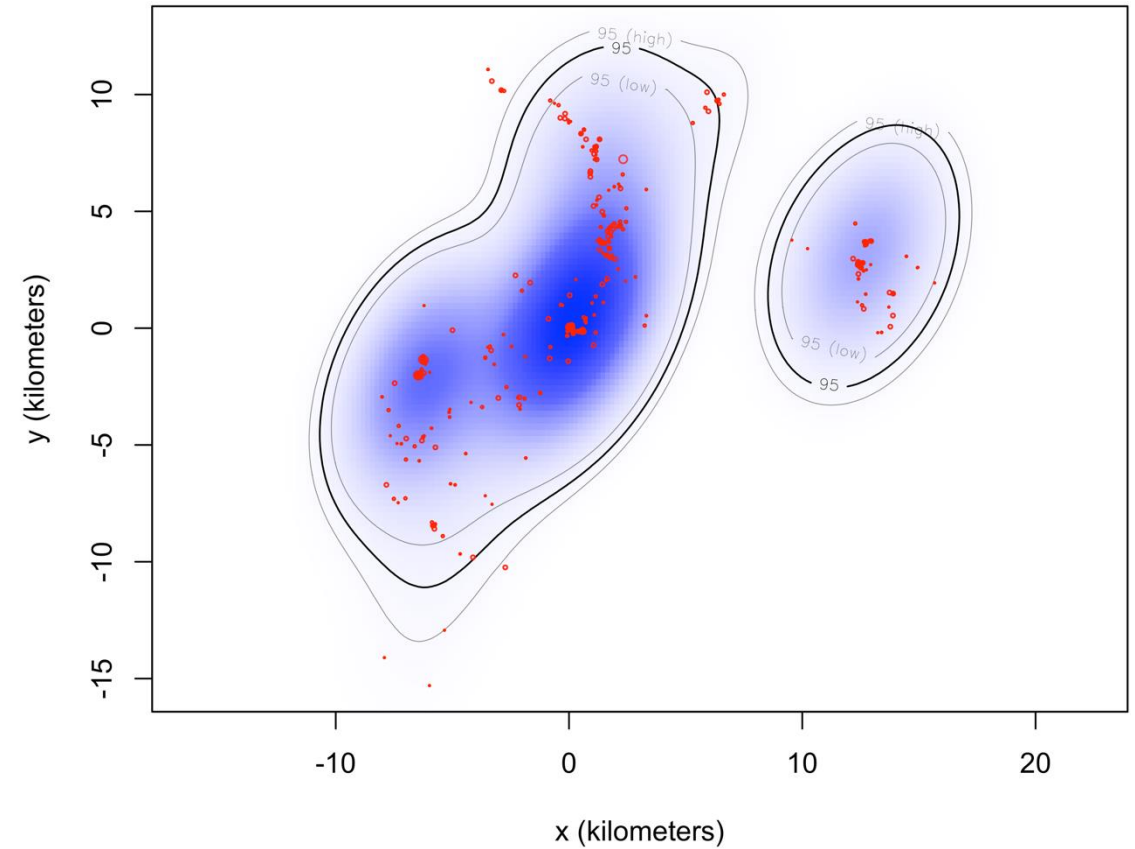
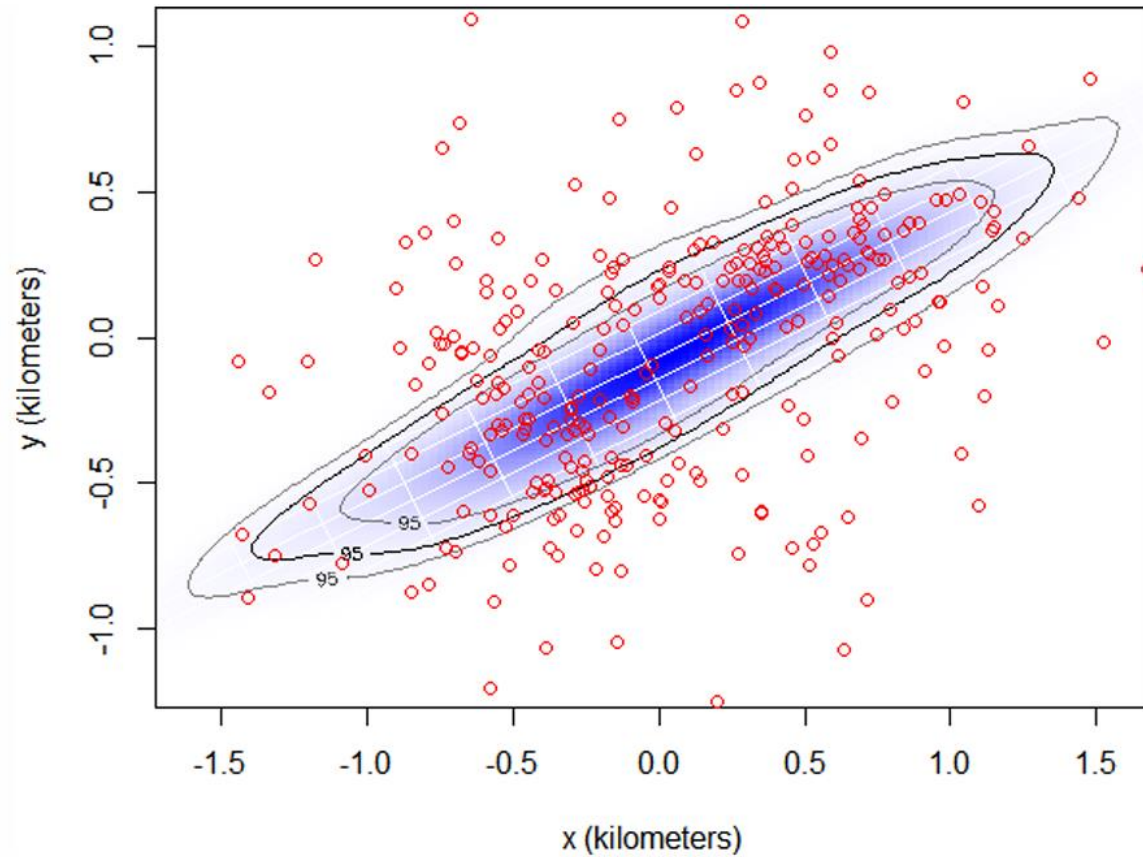
Why model error?

- Keep all of your data —data are informative
- ARGOS data or finely-sampled GPS data
 - How do the scales of movement compare to the scales of error?
- If errors are associated with a variate, then inference can become biased/incorrect

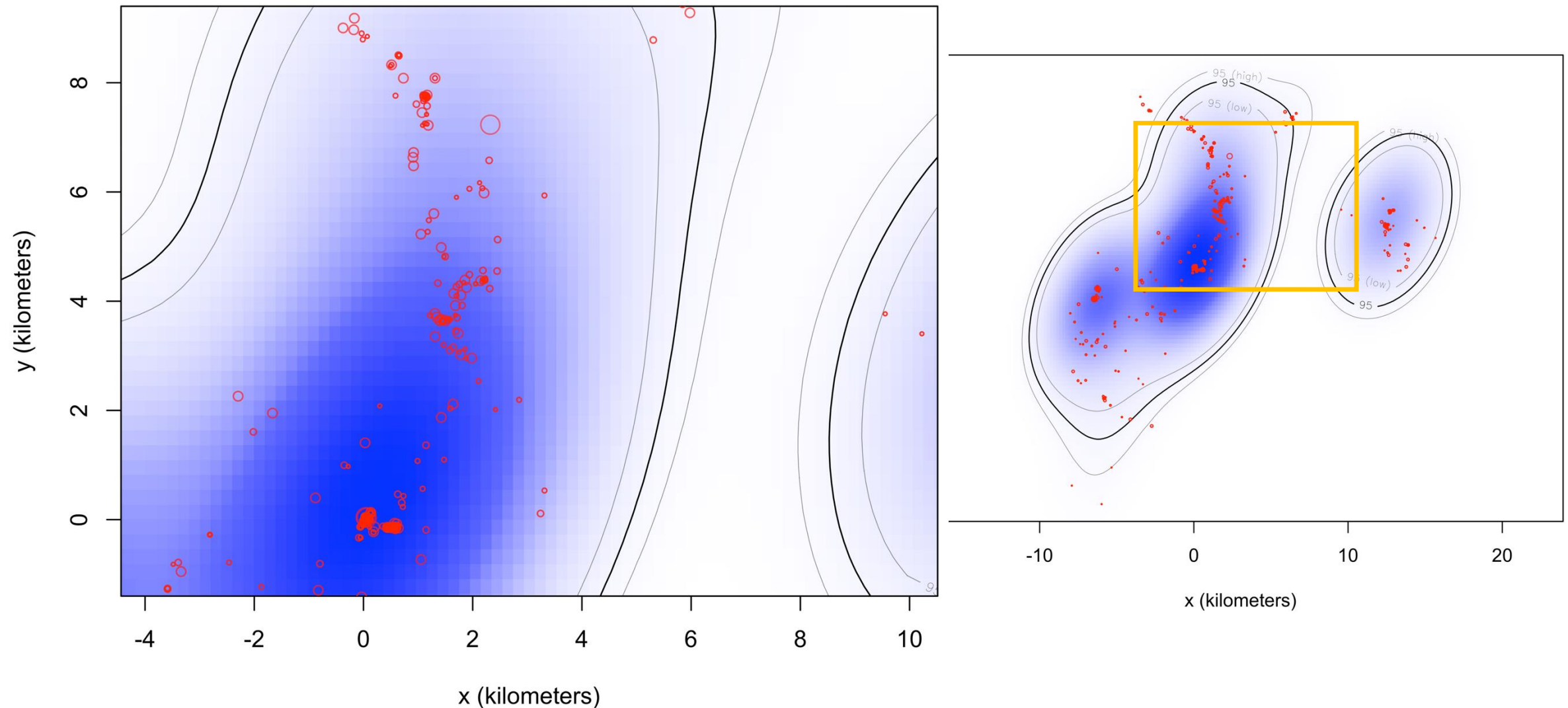
Tools available that can incorporate error

- **crawl** (2009–present)
- **ctmm** (2015–present)

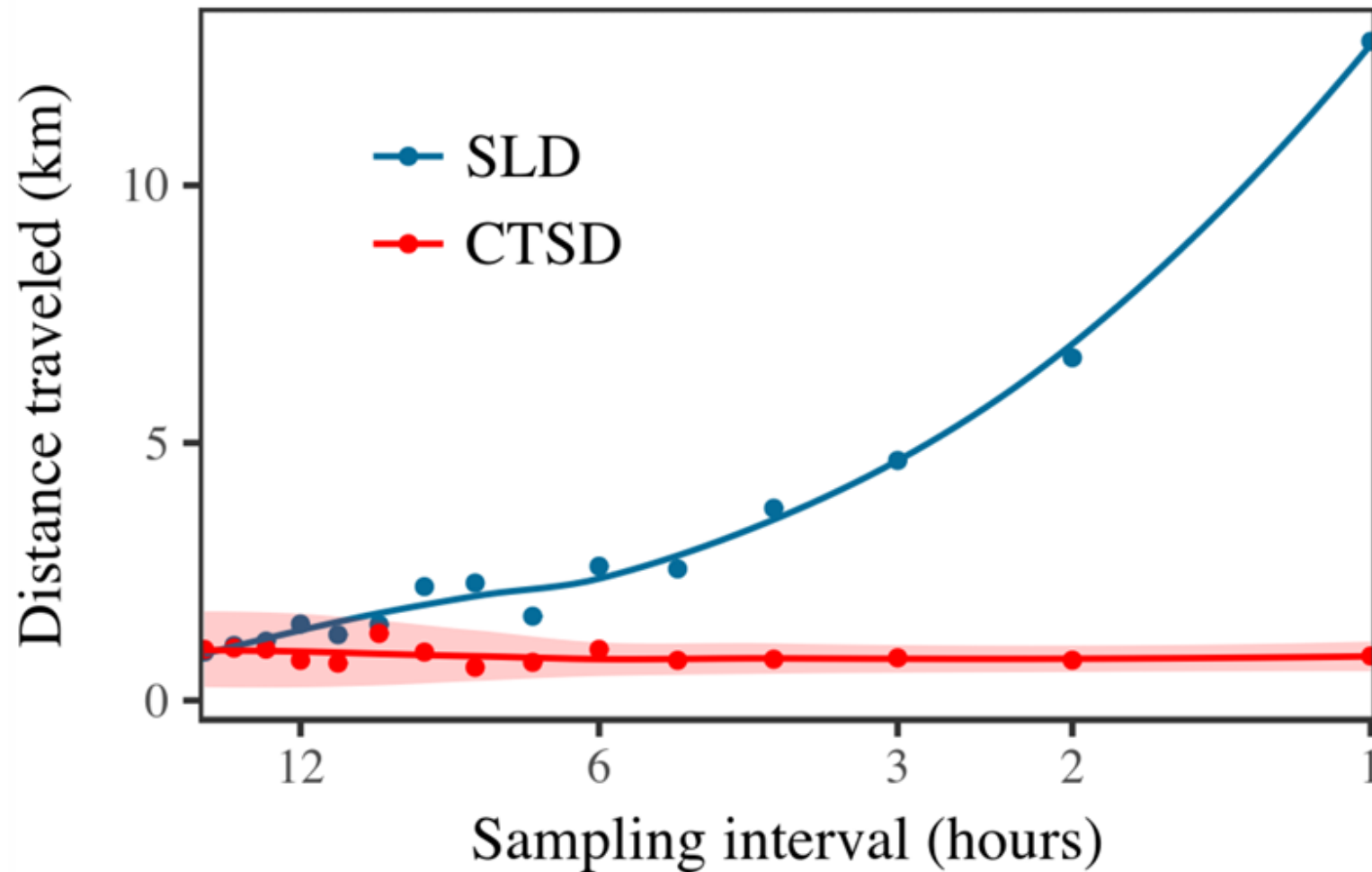
Error and home-range estimation



Error and home-range estimation



Error and speed estimation



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Estimating error parameters

- Conventional wisdom in ecology when dealing with observation error:

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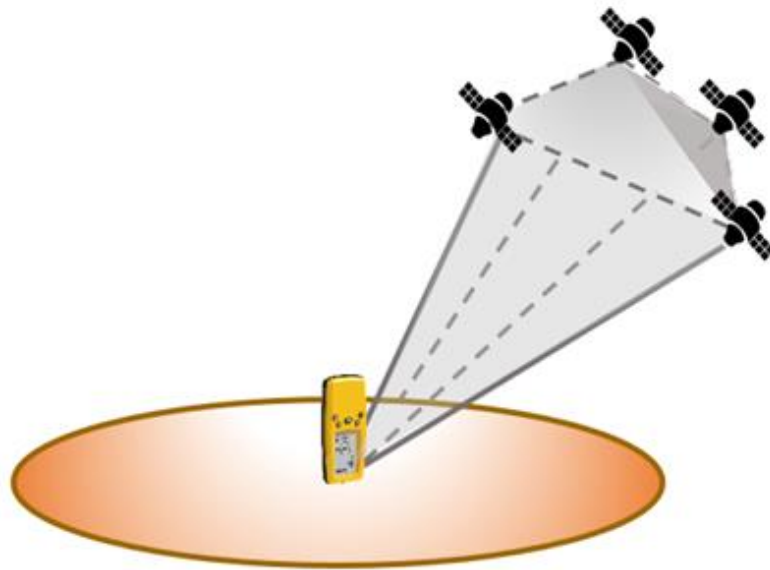
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- Solution:

Estimating error parameters

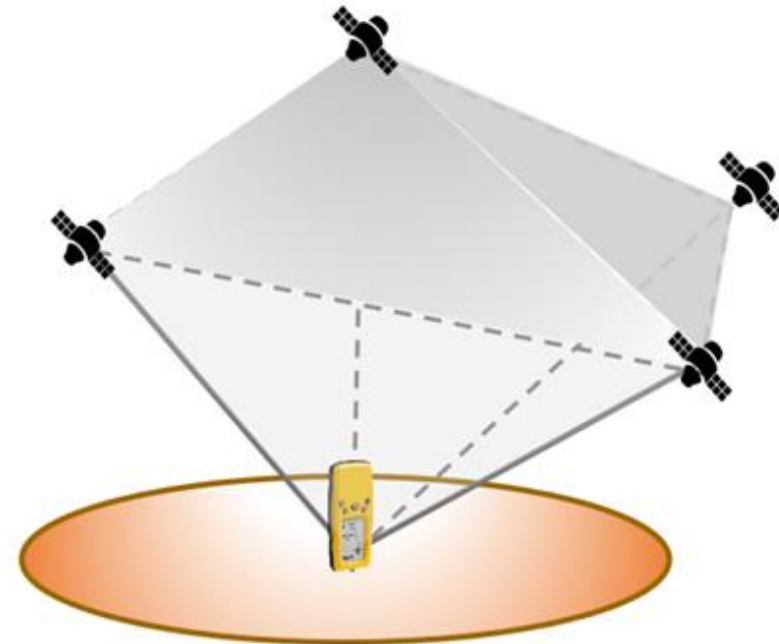
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- Solution: Error calibration

Satellite telemetry error (DOP values)

Satellite Geometry and Dilution of Precision (DOP)

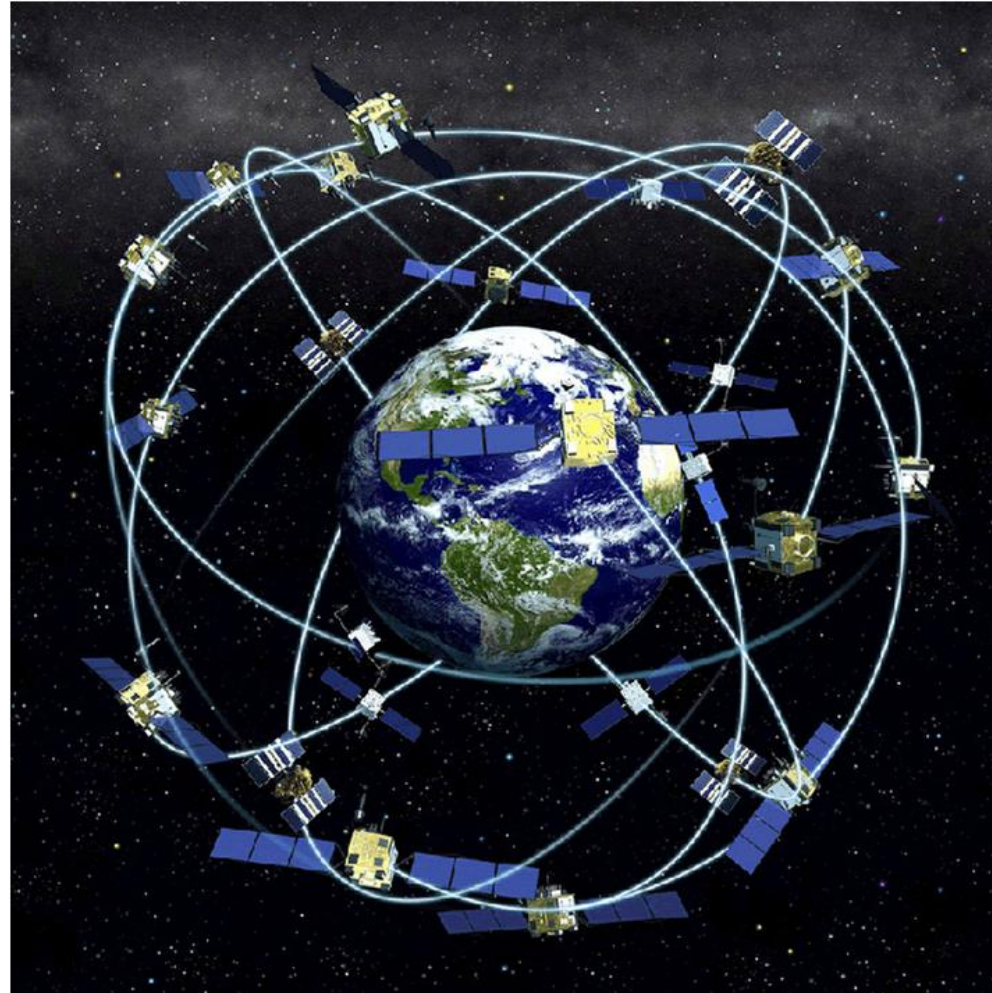


(a) Satellites clustered together
Satellite geometry enclosing less volume
BAD DOP (HIGH DOP VALUE)

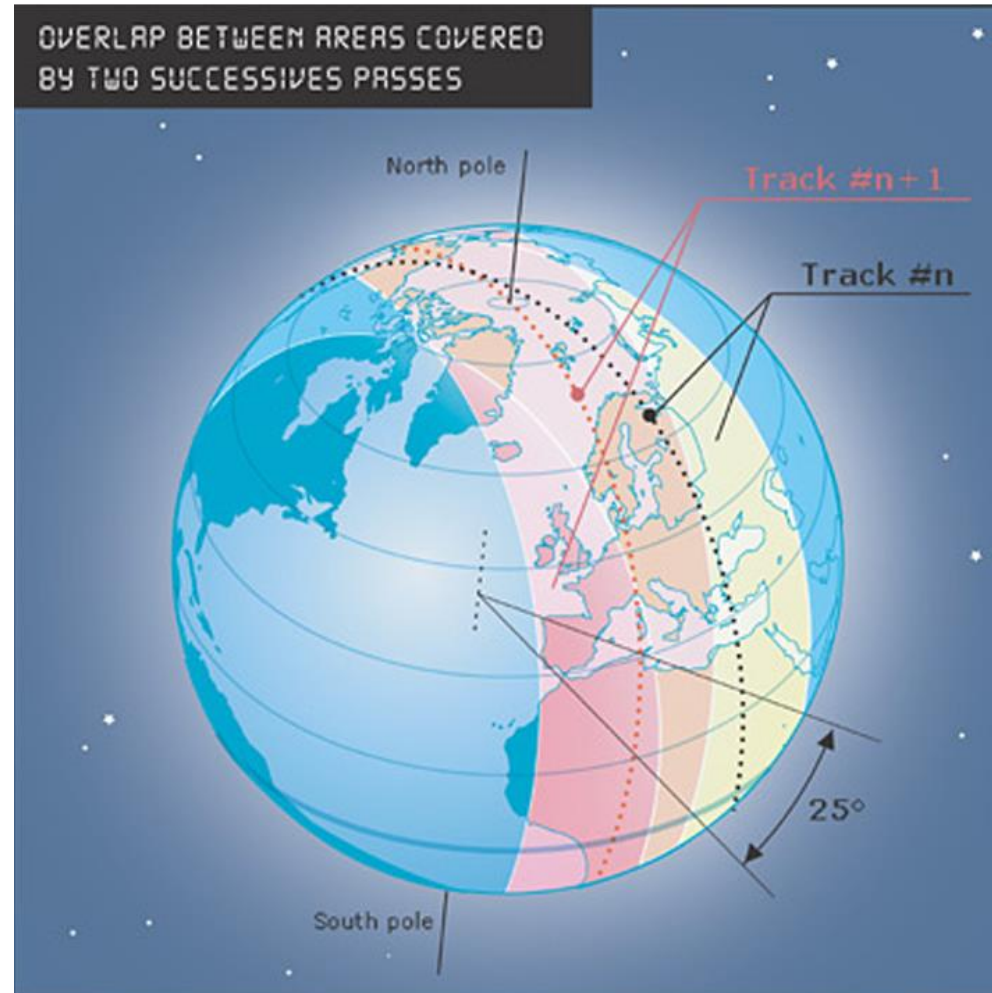


(b) Satellites far apart in the sky
Satellite geometry enclosing more volume
GOOD DOP (LOW DOP VALUE)

GPS satellite constellation



GPS satellite constellation



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- 3) HDOP & VDOP



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	t	HDOP	class	x	y
1	0	1.2	3D	-1.70300692	3.9283671
2	1200	5.3	3D	3.64006493	-2.1654782
3	1800	3.7	3D	-7.51037806	-7.2705485
4	2400	1.8	3D	-11.20956077	-6.3041996
5	3000	3.0	3D	-16.98028580	-22.2938127
6	3600	2.1	3D	2.48674965	1.3010680

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Error circle $\propto$ DOP

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- 5) Location classes (2D/3D, timed out, ...)

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- 4) Number of satellites
- 5) Location classes (2D/3D, timed out, ...)
- 6) Homoskedastic errors (models of last resort)

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- MVU AIC_C values for error models

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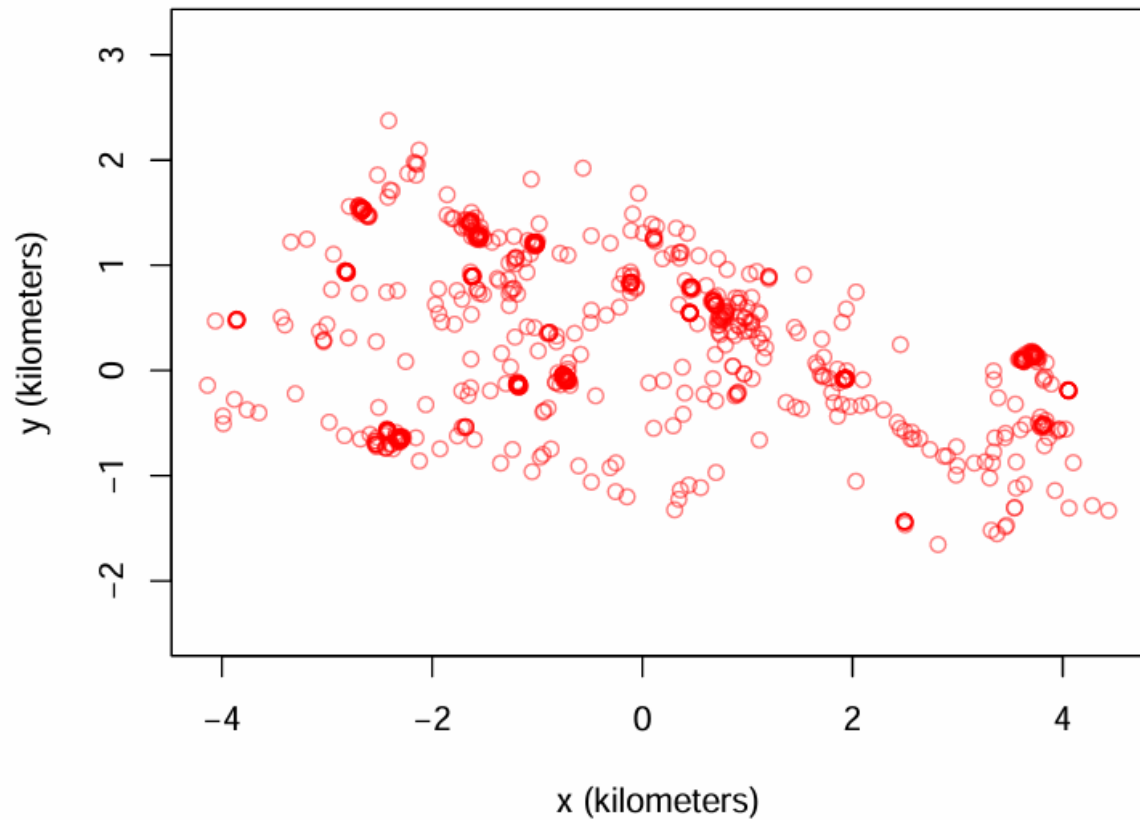
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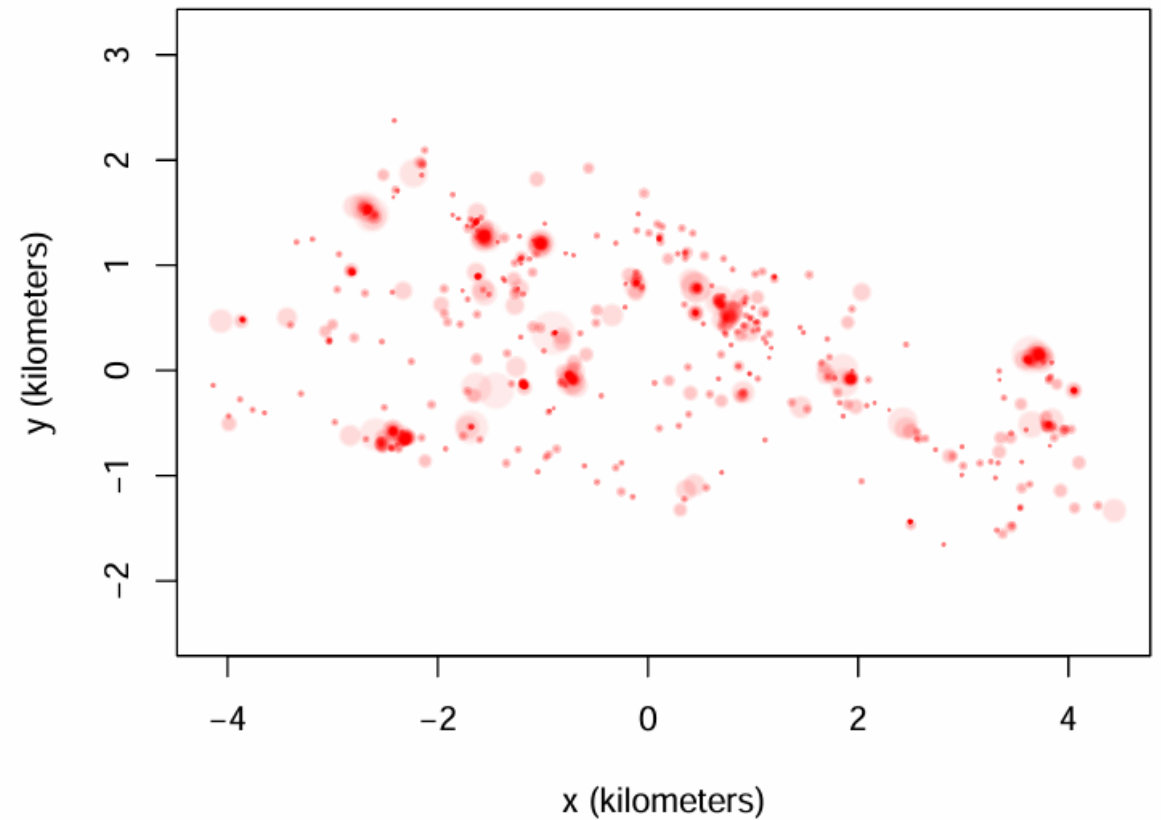
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- `vignette("error")`

What does calibrated data look like?

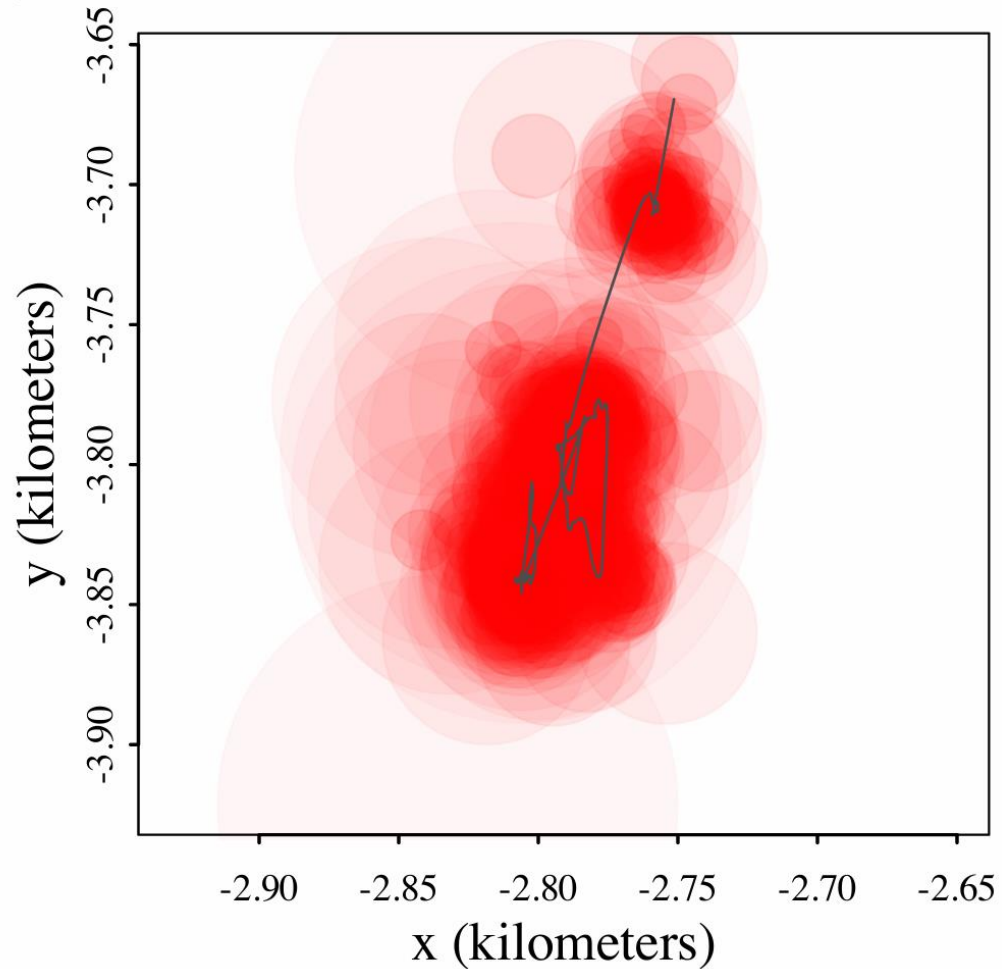
Raw data



Calibrated data



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- How large are the errors
 - How predictable is the heteroskedasticity
- This can sometimes be very important



ctmm_error.R

Thank you

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- Funded by and supported by:

